

Sequences and series - Set 2 - Solutions

Specification MM2.1, MM2.2, MM2.3, M4.19.

Answer key

Question	Answer	Topic
1	A	Sequences and series
2	F	Sequences and series
3	A	Sequences and series
4	H	Sequences and series
5	B	Sequences and series
6	B	Sequences and series
7	C	Sequences and series
8	C	Sequences and series
9	A	Sequences and series
10	E	Sequences and series
11	H	Sequences and series

Worked solutions

Question 1 SEQ-16 · Sequences and series · spec MM2.2

The sum of the first n terms of an arithmetic series is S_n . Given that $S_{10} = 145$ and $S_{20} = 490$, find the common difference.

Answer: A

$$S_{10} = 5(2a + 9d) = 145 \Rightarrow 2a + 9d = 29; S_{20} = 10(2a + 19d) = 490 \Rightarrow 2a + 19d = 49. \text{ Subtract: } 10d = 20, d = 2.$$

Full method - how and why

Two sums give two equations in a and d . Subtracting eliminates a - and only d is wanted.

1 $S_{10} = \frac{10}{2}[2a + 9d] = 145$, so $2a + 9d = 29$.

why: Divide out the $\frac{n}{2}$ factor straight away to keep the equations clean.

2 $S_{20} = \frac{20}{2}[2a + 19d] = 490$, so $2a + 19d = 49$.

why: Same formula with $n = 20$.

3 Subtract: $10d = 20$, so $d = 2$.

why: The $2a$ terms are identical in both equations, so subtraction removes them at once.

Where it goes wrong

- Using $(n - 1)d$ as $10d$ and $20d$ rather than $9d$ and $19d$.
- Dividing 490 by 20 instead of by 10.

Takeaway Two sums, two unknowns: write both in the $2a + (n - 1)d$ form and subtract to eliminate a .

Question 2 SEQ-18 · Sequences and series · spec MM2.1

A sequence is defined by $u_1 = 1$ and $u_{n+1} = u_n + 2n + 1$. Which of the following is an expression for u_n ?

Answer: F

$$u_2 = 4, u_3 = 9, u_4 = 16: u_n = n^2. \text{ Check: } (n + 1)^2 = n^2 + 2n + 1.$$

Full method - how and why

A recurrence that generates a familiar sequence. Compute terms, recognise the pattern, then verify it satisfies the rule.

1 Generate terms: $u_1 = 1, u_2 = 1 + 3 = 4, u_3 = 4 + 5 = 9, u_4 = 9 + 7 = 16$.

why: The increments 3, 5, 7 are the odd numbers - a strong hint.

2 Recognise 1, 4, 9, 16 as the squares: conjecture $u_n = n^2$.

why: Pattern-spotting is legitimate here because the options let you verify.

3 Verify against the recurrence: if $u_n = n^2$ then $u_n + 2n + 1 = n^2 + 2n + 1 = (n + 1)^2 = u_{n+1}$ ✓.

why: This is the crucial step: checking the formula satisfies the rule turns a guess into a proof.

Where it goes wrong

- Choosing $2n - 1$ (the increments) instead of the terms themselves.
- Testing only u_1 , which several options match.

Takeaway Spot the pattern from a few terms, then prove it by substituting into the recurrence.

Question 3 X2-04 · Sequences and series · spec MM2.3

A geometric series has first term $a \neq 0$ and common ratio r . P : ' $-1 < r < 1$ '. Q : 'the series has a sum to infinity'. Which one of the following is true?

Answer: A

A geometric series converges if and only if $|r| < 1$, which is exactly $-1 < r < 1$.

Full method - how and why

A definition-driven case: the convergence condition for a geometric series is exactly the stated inequality, so both directions hold.

1 A geometric series with $a \neq 0$ has a sum to infinity if and only if $|r| < 1$.

why: The partial sums $\frac{a(1-r^n)}{1-r}$ converge precisely when $r^n \rightarrow 0$.

2 $|r| < 1$ is the same statement as $-1 < r < 1$.

why: Modulus inequalities unfold into a double inequality.

3 So P and Q are equivalent: necessary **and** sufficient.

why: An 'if and only if' condition gives both directions at once.

Where it goes wrong

- Forgetting the case $r = \pm 1$: $r = 1$ gives partial sums na , which diverge; $r = -1$ oscillates. Both are correctly excluded.
- Answering 'sufficient but not necessary' from habit - here no counterexample exists.
- Overlooking why $a \neq 0$ is stated: with $a = 0$ every r would give a (trivial) sum.

Takeaway Geometric series converge $\Leftrightarrow |r| < 1$, and then $S_\infty = \frac{a}{1-r}$. An 'if and only if' condition is necessary and sufficient.

Question 4 X2-23 · Sequences and series · spec MM2.2

An arithmetic sequence has first term a and common difference d , both non-zero. Which of the following statements is/are true?

- I If $d > 0$ then every term after some point is positive.
- II The sum of the first n terms is a quadratic function of n .
- III If a and d are integers then S_n is an integer for every n .

Answer: H

I: terms $a + (n - 1)d$ eventually exceed any bound. II: $S_n = \frac{n}{2}(2a + (n - 1)d)$ is quadratic in n . III: S_n is a sum of integers.

Full method - how and why

Three claims about an arithmetic sequence, settled by the n th-term and sum formulas.

- 1 I: the terms are $a + (n - 1)d$ with $d > 0$, which increases without bound, so beyond some point every term is positive. **True.**
why: A positive common difference eventually overcomes any negative start.
- 2 II: $S_n = \frac{n}{2}(2a + (n - 1)d) = \frac{d}{2}n^2 + (a - \frac{d}{2})n$ - a quadratic in n (with $d \neq 0$, so the n^2 term survives). **True.**
why: Expand and collect powers of n ; the hypothesis $d \neq 0$ is what makes it genuinely quadratic.
- 3 III: S_n is a sum of n integers $a, a + d, \dots$, hence an integer. **True.**
why: Sums of integers are integers - no need to worry about the $\frac{n}{2}$ in the formula, since the bracket is even when n is odd and vice versa.
- 4 So all three.
why: Match to the list.

Where it goes wrong

- Rejecting III because the formula contains $\frac{n}{2}$ - the product still comes out whole.
- Rejecting II if d were 0 (linear) - but the question states $d \neq 0$.
- Reading I as 'every term is positive' rather than 'every term after some point'.

Takeaway S_n for an arithmetic series is quadratic in n ; check the stated conditions (a, d non-zero) before dismissing a claim.

Question 5 SEQ-05 · Sequences and series · spec MM2.3

A geometric series has first term a and common ratio r with $|r| < 1$. Its sum to infinity is 8 and the sum to infinity of the squares of its terms is $\frac{64}{3}$. Find r .

Answer: B

$$\frac{a}{1-r} = 8 \text{ and } \frac{a^2}{1-r^2} = \frac{64}{3}. \text{ Dividing the second by the square of the first: } \frac{1-r}{1+r} = \frac{64/3}{64} = \frac{1}{3}, \text{ so } 3-3r=1+r \text{ and } r=\frac{1}{2}.$$

Full method - how and why

Two infinite sums, two unknowns. Dividing one equation by the (square of the) other eliminates a elegantly.

1 Write the two facts: $\frac{a}{1-r} = 8$ and, for the squares (first term a^2 , ratio r^2), $\frac{a^2}{1-r^2} = \frac{64}{3}$.
 why: Squaring every term of a geometric series gives another geometric series with first term a^2 and ratio r^2 .

2 Square the first: $\frac{a^2}{(1-r)^2} = 64$.
 why: Squaring makes the numerators match, so they can be eliminated by division.

3 Divide the second equation by this: $\frac{(1-r)^2}{1-r^2} = \frac{64/3}{64} = \frac{1}{3}$.
 why: Dividing kills a^2 entirely - the standard move when two equations share a factor.

4 Simplify using $1-r^2 = (1-r)(1+r)$: $\frac{1-r}{1+r} = \frac{1}{3}$.
 why: One factor of $(1-r)$ cancels.

5 Cross-multiply: $3-3r=1+r \Rightarrow 4r=2 \Rightarrow r=\frac{1}{2}$.
 why: Check: $a=4$, and $\frac{16}{1-1/4} = \frac{64}{3}$ ✓.

Where it goes wrong

- Using ratio r (not r^2) for the series of squares.
- Forgetting to square the first equation before dividing, which leaves an a behind.

Takeaway Squaring a geometric series squares both a and r ; divide equations to eliminate the unknown you do not want.

Question 6 SEQ-07 · Sequences and series · spec MM2.1

A sequence satisfies $x_{n+1} = \frac{1}{1-x_n}$ with $x_1 = 2$. Find x_{100} .

Answer: B

$$x_2 = \frac{1}{1-2} = -1, x_3 = \frac{1}{1-(-1)} = \frac{1}{2}, x_4 = \frac{1}{1-1/2} = 2 = x_1: \text{ the sequence has period 3. Since } 100 = 3 \times 33 + 1, x_{100} = x_1 = 2.$$

Full method - how and why

A recurrence with a huge index. Compute a few terms, spot the cycle, then use arithmetic on the index.

1 Iterate: $x_2 = \frac{1}{1-2} = -1, x_3 = \frac{1}{1-(-1)} = \frac{1}{2}, x_4 = \frac{1}{1-1/2} = 2$.
 why: When an index like 100 appears, always compute the first few terms - the sequence is almost certainly periodic.

2 $x_4 = x_1$, so the sequence repeats with period 3: $2, -1, \frac{1}{2}, 2, -1, \frac{1}{2}, \dots$
 why: Once a value repeats, the whole sequence from that point repeats identically.

3 Reduce the index: $100 = 3 \times 33 + 1$, so x_{100} is in the same position of the cycle as x_1 .
 why: Terms with indices differing by a multiple of the period are equal; compare remainders on division by 3.

4 Therefore $x_{100} = x_1 = 2$.
 why: Check the convention: the indices 1, 4, 7, ... all leave remainder 1 on division by 3, and 100 does too.

Where it goes wrong

- Finding that 100 leaves remainder 1 but then reading off x_0 or x_3 - line the remainders up with the actual starting index.
- Giving up after two terms and guessing -1 or $\frac{1}{2}$.

Takeaway Large-index recurrence questions are periodicity questions: find the cycle length, then work with the index modulo it.

Question 7 SEQ-11 · Sequences and series · spec MM2.2

The sum of the first n natural numbers is S . The sum of the first $2n$ natural numbers is kS . Which of the following is true?

Answer: C

$$S = \frac{n(n+1)}{2} \text{ and the sum to } 2n \text{ is } \frac{2n(2n+1)}{2} = n(2n+1). \text{ So } k = \frac{n(2n+1)}{n(n+1)/2} = \frac{2(2n+1)}{n+1}.$$

Full method - how and why

A ratio of two triangular numbers. The answer depends on n , so the constant options can be eliminated immediately.

- Write both sums: $S = \frac{n(n+1)}{2}$ and the sum to $2n$ is $\frac{2n(2n+1)}{2} = n(2n+1)$.
why: Use $1 + 2 + \dots + m = \frac{m(m+1)}{2}$ with $m = n$ and $m = 2n$.
- Form the ratio: $k = \frac{n(2n+1)}{\frac{n(n+1)}{2}}$.
why: k is defined by 'sum to $2n$ ' = $k \times$ 'sum to n '.
- Simplify: multiply top and bottom by 2 and cancel n : $k = \frac{2n(2n+1)}{n(n+1)} = \frac{2(2n+1)}{n+1}$.
why: Cancelling n is legitimate since $n \geq 1$.
- Sanity check with $n = 1$: sums are 1 and 3, so $k = 3$; the formula gives $\frac{2 \times 3}{2} = 3$ ✓.
why: A single numerical check rules out all the other options at once.

Where it goes wrong

- Answering $k = 4$ because 'doubling n roughly quadruples a quadratic' - true in the limit, false for finite n .
- Forgetting the factor 2 when dividing by a fraction.

Takeaway When the options are a mix of constants and formulas, substitute a small value of n - it eliminates most of them instantly.

Question 8 SEQ-17 · Sequences and series · spec MM2.3

A ball is dropped from a height of 10 m. After each bounce it rises to $\frac{3}{5}$ of its previous height. Find the total distance travelled by the ball before it comes to rest.

Answer: C

$$\text{Down: } 10 + 6 + 3.6 + \dots = \frac{10}{1 - 3/5} = 25. \text{ Up: } 6 + 3.6 + \dots = 15. \text{ Total } 40 \text{ m.}$$

Full method - how and why

A classic bouncing-ball problem: the ball goes down **and** up. Two geometric series, or one with a correction.

- Downward distances: 10, 6, 3.6, ... - geometric with $a = 10$, $r = \frac{3}{5}$. Sum = $\frac{10}{1 - \frac{3}{5}} = 25$.
why: After each bounce it falls from the new height, so every rise is matched by an equal fall.
- Upward distances: 6, 3.6, ... - the same series without the initial 10. Sum = $25 - 10 = 15$.
why: The ball never rises 10 m: the first rise is 6 m, so the up-series starts one term later.
- Total: $25 + 15 = 40$ m.
why: Add the two, or use $2 \times 25 - 10 = 40$ (double everything, then remove the un-matched first drop).

Where it goes wrong

- Answering 25 by counting only the downward distances.
- Answering 50 by doubling without subtracting the initial drop.

Takeaway Distance travelled counts every metre in both directions: total = $2S - (\text{first drop})$.

Question 9 SEQ-19 · Sequences and series · spec MM2.3

The first three terms of a geometric sequence are x , $x + 6$ and $x + 15$. Find the fourth term.

Answer: A

Geometric: $(x + 6)^2 = x(x + 15) \Rightarrow x^2 + 12x + 36 = x^2 + 15x \Rightarrow x = 12$. The terms are 12, 18, 27 with ratio $\frac{3}{2}$, so the fourth term is $27 \times \frac{3}{2} = \frac{81}{2}$.

Full method - how and why

Three consecutive geometric terms: the middle one squared equals the product of its neighbours. That single equation finds x .

- 1 Use the geometric condition $\frac{x+6}{x} = \frac{x+15}{x+6}$, i.e. $(x+6)^2 = x(x+15)$.
why: Consecutive terms have a constant ratio, so equating the two ratios gives the condition $b^2 = ac$.
- 2 Expand: $x^2 + 12x + 36 = x^2 + 15x$.
why: The x^2 terms cancel, leaving a linear equation - which is why this method is quick.
- 3 Solve: $36 = 3x$, so $x = 12$.
why: The terms are 12, 18, 27.
- 4 Find $r = \frac{18}{12} = \frac{3}{2}$ and the fourth term $27 \times \frac{3}{2} = \frac{81}{2}$.
why: The ratio need not be an integer; leave the answer as an improper fraction.

Where it goes wrong

- Assuming the common ratio is $\frac{x+6}{x+15}$ (upside down).
- Stopping at the third term 27 or giving the ratio instead of the fourth term.

Takeaway Three terms in geometric progression satisfy $b^2 = ac$; the x^2 terms always cancel, leaving a linear equation.

Question 10 SEQ-20 · Sequences and series · spec MM2.2

The integers from 1 to 100 inclusive are written down. What is the sum of those that are multiples of 3 or 5 (or both)?

Answer: E

Multiples of 3: $3 + \dots + 99 = 3(1 + \dots + 33) = 3 \times 561 = 1683$. Multiples of 5: $5(1 + \dots + 20) = 1050$. Multiples of 15: $15(1 + \dots + 6) = 315$. Total $1683 + 1050 - 315 = 2418$.

Full method - how and why

A counting-and-summing problem needing inclusion-exclusion: multiples of 15 belong to both lists and would otherwise be counted twice.

- 1 Multiples of 3 up to 100: $3 + 6 + \dots + 99 = 3(1 + 2 + \dots + 33) = 3 \times \frac{33 \times 34}{2} = 3 \times 561 = 1683$.
why: Factor out the 3 so the bracket is a plain triangular number.
- 2 Multiples of 5: $5(1 + \dots + 20) = 5 \times 210 = 1050$.
why: $\lfloor 100/5 \rfloor = 20$ terms.
- 3 Multiples of both, i.e. of 15: $15(1 + \dots + 6) = 15 \times 21 = 315$.
why: 'Or' in mathematics includes 'both', and multiples of 15 have been counted in each of the first two sums.
- 4 Inclusion-exclusion: $1683 + 1050 - 315 = 2418$.
why: Add the two sets, subtract the overlap once.

Where it goes wrong

- Forgetting to subtract the multiples of 15, giving 2733 - which is offered.
- Using $\text{lcm}(3, 5) = 8$ or 30 instead of 15.

Takeaway For 'multiples of a or b ', subtract the multiples of $\text{lcm}(a, b)$ once.

Question 11 SEQ-21 · Sequences and series · spec MM2.1

A sequence has $u_1 = 3$ and $u_{n+1} = \frac{u_n + 5}{2}$. Which of the following statements is/are true?

I The sequence is increasing.

II $u_n < 5$ for all n .

III $u_n = 5 - 2^{2-n}$ for all n .

Answer: H

$u_2 = 4$, $u_3 = 4.5$, $u_4 = 4.75$. If $u_n < 5$ then $u_{n+1} = \frac{u_n + 5}{2} < 5$ and $u_{n+1} - u_n = \frac{5 - u_n}{2} > 0$, so I and II hold. III: $5 - 2^{2-n}$ gives 3, 4, 4.5, ... and satisfies the recurrence: $\frac{5 - 2^{2-n} + 5}{2} = 5 - 2^{1-n}$. All three are true.

Full method - how and why

A recurrence converging to a limit. Compute terms for intuition, then prove the two general claims and verify the closed form.

1 Generate terms: $u_1 = 3$, $u_2 = \frac{3+5}{2} = 4$, $u_3 = 4.5$, $u_4 = 4.75$ - rising towards 5.

why: The rule averages u_n with 5, so each term sits halfway between the previous one and 5.

2 II: show $u_n < 5$ always. If $u_n < 5$ then $u_{n+1} = \frac{u_n + 5}{2} < \frac{5+5}{2} = 5$; and $u_1 = 3 < 5$. **True.**

why: This is an induction argument in miniature: true at the start, and preserved by the rule.

3 I: $u_{n+1} - u_n = \frac{u_n + 5}{2} - u_n = \frac{5 - u_n}{2} > 0$ because $u_n < 5$. So the sequence increases. **True.**

why: Compute the difference algebraically rather than relying on the first few terms; it also shows *why* it increases - the gap to 5 is always positive.

4 III: test $5 - 2^{2-n}$: at $n = 1$ it gives $5 - 2 = 3$ ✓, and $\frac{(5 - 2^{2-n}) + 5}{2} = 5 - 2^{1-n}$, which is the formula at $n + 1$ ✓. **True.**

why: Check the starting value **and** that the formula satisfies the recurrence - both are needed.

5 All three are true.

why: Match to the options.

Where it goes wrong

- Concluding II from the first four terms alone - the general argument is what makes it certain.
- Checking III only at $n = 1$; many formulas share a first term.

Takeaway To prove a claim about every term: check the first case, then show the recurrence preserves the property.